

Explainability and Error Analysis

Phase 6 — What the Model's Mistakes Are, and Whose Standard to Judge Them By

Table 1. Pre-registered verdicts, selected by the frozen rules in reports/phase6_prereg.json.

Pre-registered endpoint	Verdict
P6-A human comparator (contested), pre-registered rule	ABOVE THE HUMAN PANEL
P6-A what may actually be claimed (P6-AMD-5)	OUT-PREDICTS AN INDIVIDUAL ANNOTATOR, BUT NOT THE PANEL'S OWN MODAL VOTE.
P6-B confusion geometry — wall axis	MIRRORS HUMAN GEOMETRY
P6-B confusion geometry — station axis	DIVERGES FROM HUMAN GEOMETRY (model less adjacent)
P6-C1 attribution vs human disagreement	NOT ESTIMABLE (vote entropy is constant on this stratum by construction)
P6-C1b same, exploratory substitute signal	NO ASSOCIATION
P6-C2 attribution stability	ATTRIBUTION DESTABILISES ON CONTESTED IMAGES
P6-D selective prediction vs the Phase 5 ranking	CONSISTENT WITH THE PHASE 5 RANKING

Pre-registration frozen 2026-07-28 23:21:35 at commit a049a701, before any endpoint was scored.

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Abstract

Background. Phases 3 to 5 established one durable result and three unresolved ones. The durable result is a calibration collapse across strata of expert agreement that no target construction repairs. The unresolved ones — accuracy, uncertainty and the anatomy-aware loss — are all comparisons between models. None of them can say whether the model's residual error reflects genuine visual ambiguity or a limit of the model, because no comparator outside the model set was ever measured.

Objective. To change the comparator. Four pre-registered endpoints: whether the model is distinguishable from a held-out human annotator scored identically (P6-A); whether its error geometry mirrors human disagreement geometry when both sides carry intervals (P6-B); whether spatial attribution tracks human disagreement where predictive entropy did not (P6-C); and whether the Phase 5 out-of-protocol rejection ranking survives a threshold-free treatment (P6-D).

Methods. No retraining. The frozen checkpoints of all five arms across three seeds were evaluated on the same 1,353-image test split in the same row order Phases 3 and 4 used, verified by reproduction gates that recover the published Phase 3 numbers exactly. Every internal interval is a patient-clustered bootstrap of 1,000 resamples; contrasts are paired, with both sides scored on the same resample before differencing.

Results. On the contested strata the model out-predicts a held-out annotator by the pre-registered rule ($\Delta = 0.0647$ macro F1, 95% CI [0.028, 0.100]). A post-hoc sensitivity analysis qualifies that sharply: the modal vote of the same three references scores 0.6700 against 0.3902 for the annotator, and the model reaches only 24% of that headroom. The model therefore beats an individual expert but not the panel's own aggregation rule, and a substantial model shortfall against the attainable ceiling remains. Its wall-confusion geometry mirrors the human geometry, while its station geometry diverges by -8.19 points (CI [-15.10, -2.85]). Selective prediction separates the arms far more sharply externally (AURC 0.1906 for C2 against 0.3235 for C3) than internally, and agrees with the Phase 5 rejection ordering.

Conclusion. Two things degrade as annotators disagree, and the literature reports them as one. The attainable ceiling falls — the best single-label predictor drops from 1.00 to 0.67 — which accounts for much of the decline Phase 3 attributed to the model. But a real model shortfall against that reduced ceiling remains, so the decline is not purely a property of the task either. What is unambiguous is that confidence degrades further and faster than discrimination, that no target construction repairs it, and that the endpoint which exposes it is the model's willingness to decline rather than its accuracy.

1. Introduction

1.1 What Phases 3–5 settled, and the three debts they left

Phase 3 measured performance across strata of annotator agreement and found annotator-marginalized macro F1 falling from 83.92 on unanimous images to 26.15 on the 2-1-1 stratum, with expected calibration error rising from 9.15% to 56.40%. Phase 4 tested five target constructions and found none that repaired the calibration. Phase 5 carried all five arms to two external corpora and found that the label spaces could not express the wall $\times$ station grid at all, reframing the endpoint before scoring. Three debts were left explicitly open.

1.1.1 X3: a withdrawn claim about confusion geometry

The debt recorded as X3

Phase 3 reported that the model's wall-confusion geometry (89.68%) sat within 0.12 points of the human value (89.8%) and that its station-confusion geometry (85.57%) trailed the human 93.1% by 7.5 points. Correction X3 withdrew both to hypothesis status: the model shares carried patient-clustered intervals, the human values were corpus-wide point estimates with none, and the station interval contained the human value. A comparison between an interval and a point is not a comparison.

1.1.2 The blueprint's question: ambiguity or capacity?

The blueprint asks of this phase whether "the residual error reflects genuine visual ambiguity rather than model capacity". Confusion geometry alone cannot settle that. It shows that model errors have the same shape as human disagreements; it does not show that the images are equally hard for both. Only a comparator drawn from the annotators themselves can license the ambiguity claim, and no such comparator existed anywhere in this project before Phase 6.

1.1.3 The Phase 5 carry-forward on out-of-protocol rejection

Phase 5 recorded that out-of-protocol rejection "separated the arms where every internal endpoint failed to" and instructed Phases 6–7 to treat it as a primary endpoint. Phase 5 measured it at exactly one operating point — whether the 23-way argmax lands on OTHERCLASS — which is a property of where a decision boundary happens to fall rather than of how well the model's confidence orders its own mistakes.

1.2 Why the comparator changes in this phase

Every comparison in Phases 3 to 5 was model against model. That design can rank target constructions but cannot calibrate the ranking against anything outside it: a model scoring 26.15 macro F1 looks broken until one knows what a board-certified endoscopist scores on the same images. Phase 6 therefore introduces two comparators the project has never used — the annotators, and the model's own confidence ordering — and keeps the explainability analysis quantitative so that it can fail.

1.3 Endpoints and pre-registered verdict rules

Table 2. The four Phase 6 endpoints, as frozen before any scoring ran.

ID	Endpoint	Question
P6 -A	human comparat or	Is the model's per-stratum performance distinguishable from that of a held-out human annotator scored the same way?
P6 -B	confusion geometry, both sides given intervals	Does the model's error geometry mirror the human disagreement geometry measured in Phase 0?

ID	Endpoint	Question
P6 -C	attribution as a measurement	Does spatial attribution track human disagreement where predictive entropy did not?
P6 -D	selective prediction	Does the Phase 5 out-of-protocol rejection ranking survive a threshold-free risk-coverage treatment?

2. Methods

2.1 What is held fixed

No retraining, no threshold tuning and no checkpoint reselection. The same frozen checkpoints, the same 224×224 cache, the same training-set normalisation and the same 1,353-image test split in the same row order that Phases 3 and 4 used. Two reproduction gates enforce this rather than assert it.

Table 3. Phase-boundary reproduction gates. Each recovers a number published in an earlier phase from the new code path.

Gate	What it checks	Result
P6 .1 a	panel row order identical to the Phase 3 cache index	PASS
P6 .1 b	C0 re-scored reproduces the Phase 3 macro F1 per stratum	PASS (worst $ \Delta = 0.0e+00$)
P6 .2 a	the 4-annotator-marginalized score equals the Phase 3 value	PASS (worst $ \Delta = 0.0e+00$)
P6 .3 a	C0's S-unanimous geometry reproduces Phase 3 exactly	PASS (89.68 / 85.57)
P6 .4 b	each CAM targets the committed prediction	PASS for 15 checkpoints
P6 .6 a	risk at full coverage equals 1 – accuracy	PASS

2.2 P6-A: the held-out-annotator construction

For each held-out annotator a in $\{0,1,2,3\}$: $\text{human_a} = \text{mean over } b \neq a \text{ of } \text{macroF1}(\text{votes[:,b]}, \text{votes[:,a]})$; $\text{model_a} = \text{mean over } b \neq a \text{ of } \text{macroF1}(\text{votes[:,b]}, \text{model_pred})$; $\text{delta_a} = \text{model_a} - \text{human_a}$. Reported as the mean over the four held-out annotators, per stratum.

Why annotator a is scored against the other three, not all four

Scoring an annotator against a reference set that contains themselves makes one of the terms an identity and inflates the human side. Excluding a from the reference costs precision and buys a comparison that is not rigged. The model is scored against the SAME three references so the two sides differ only in who produced the prediction.

2.3 P6-B: giving the human benchmark an interval

wall adjacency and station neighbouring are taken unchanged from `phase4_structure.py` / `phase3_confusion.py`: WALL_CYCLE G-A-L-P-G with cyclic distance 1 = adjacent, and $|\text{delta station}| == 1$ = neighbouring. OTHERCLASS has no geometry and is excluded from both shares.

Table 4. Event definitions. Both sides use the identical adjacency relations.

Side	What counts as one event
model	prediction differing from an annotator's label, marginalized over all four annotators
human	a disagreeing annotator pair; all 6 pairs per image considered
excluded	events involving OTHERCLASS, which has no position on the (wall x station) grid

2.4 P6-C: Grad-CAM as a scored quantity

Layer: model.features[-1] (final ConvNeXt stage output). Target: the model's own predicted class, NOT the reference label, so the map explains what the model did rather than what it should have done. Top-q for the overlap masks: 0.2, fixed a priori; no post-hoc selection of q is permitted.

Table 5. The three scored attribution quantities, fixed before any map was rendered.

Quantity	Definition
dispersion	normalised Shannon entropy of the CAM treated as a spatial probability distribution after ReLU and sum-normalisation; 1.0 = uniform over the frame, 0.0 = all mass on one cell
inter_seed_iou	mean pairwise IoU of the top-q attribution masks across the three seeds of one arm, per image
inter_arm_iou	mean pairwise IoU across arms at fixed seed, per image

Why this phase scores attribution instead of displaying it

Attribution maps are the standard way this chapter is written, and the standard way is not a measurement: a grid of heatmaps invites the reader to agree with the author. The pre-registration therefore fixed the layer, the target class and the overlap threshold in advance and committed to reporting the numbers whatever they said. Qualitative panels appear in this report only after the quantitative endpoint has been stated, and their caption carries that result.

A numerical fact the gate exposed

the CAM targets the class recorded in the committed prediction file, not this pass's argmax. Phases 3-5 ran inference under float16 autocast; this pass runs in float32. On images where the top two classes are near-tied (the smallest observed top1-top2 margin in the frozen posteriors is 3.08e-03) the two precisions can disagree about the argmax. Explaining the live argmax would then explain a decision no other number in the thesis was computed from. The divergence count is recorded per checkpoint below as a numerical fact about the evaluation path rather than suppressed.

2.5 P6-D: risk–coverage and AURC

Confidence score: max softmax probability. the simplest and most widely reported confidence score, chosen so the endpoint measures the deployed model rather than a bolt-on detector. Predictive entropy and the top-two margin are computed and tabulated as declared secondaries.

Order the images by confidence, accept the most confident fraction c , and record the error rate among the accepted. Sweeping c traces the risk–coverage curve; AURC is the area beneath it, the expected error of a model permitted to abstain, averaged over every abstention budget. A model whose confidence perfectly ordered its own mistakes would push all error into the low-confidence tail and score near zero.

3. Results

3.1 P6-A: model versus the human panel

Table 6. P6-A. Held-out annotator and model, scored by the same metric against the same three-annotator reference panel, on the same images and the same patient resamples. Headline arm C2 vote proportions.

Stratum	n	Human 95% CI	Model (C2) 95% CI	Δ	Δ 95% CI	Verdict
S-unanimous	803	[1.000, 1.000]	[0.831, 0.881]	-0.149	[-0.169, -0.119]	UNINFORMATIVE (BY CONSTRUCTION)
S-majority	342	[0.470, 0.492]	[0.477, 0.568]	+0.007	[-0.005, 0.084]	INDISTINGUISHABLE FROM THE HUMAN PANEL
S-plurality	127	[0.117, 0.144]	[0.205, 0.279]	+0.088	[0.075, 0.147]	ABOVE THE HUMAN PANEL
S-no-majority	81	[0.198, 0.246]	[0.233, 0.310]	+0.035	[0.006, 0.090]	ABOVE THE HUMAN PANEL
S-contested (pooled)	550	[0.366, 0.407]	[0.412, 0.490]	+0.044	[0.028, 0.100]	ABOVE THE HUMAN PANEL

Declared degeneracy, detected not assumed

P6-AMD-1: the pre-registration did not anticipate that the human side of P6-A is 1.0 by construction on any stratum defined by unanimity. The S-unanimous verdict is therefore reported as UNINFORMATIVE (BY CONSTRUCTION) rather than as BELOW THE HUMAN PANEL. The contested strata, where annotators genuinely disagree, are unaffected and carry the endpoint.

P6-AMD-5 — two asymmetries found after scoring

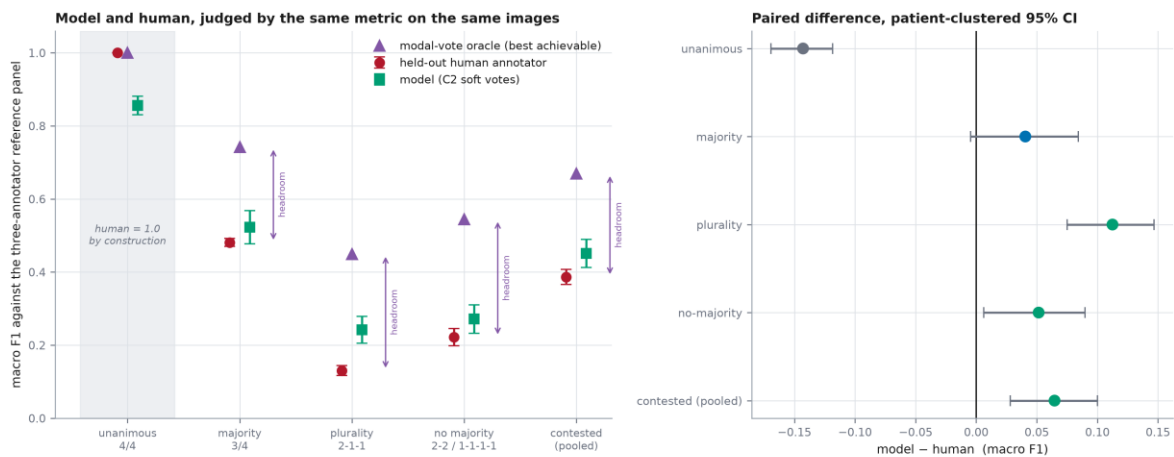
two asymmetries in the P6-A construction, found after scoring and measured rather than argued: the model is optimised to predict THIS panel's consensus and the held-out annotator is not (exposure); and the model chooses a label while the held-out annotator is stuck with the one they gave (choice). the pre-registered verdict 'ABOVE THE HUMAN PANEL' invites the reading that the classifier out-performs a board-certified endoscopist. The measurements below do not support that reading in general. They support a narrower and more interesting one, stated per stratum.

Table 7. P6-AMD-5. The modal vote of the same three references is the best any single-label predictor can achieve against them, so it bounds the comparison. The singleton rate is the fraction of held-out annotators whose label is shared by none of the other three — a structural handicap fixed by the stratum definition, not by skill.

Stratum	Held-out human	Model (C2)	Modal-vote oracle	Headroom recovered	Single ton rate	Exceeds oracle?
S-unanimous	1.0000	0.8514	1.0000	n/a	0%	no
S-majority	0.4893	0.5301	0.7423	16%	25%	no
S-plurality	0.1436	0.2668	0.4492	40%	50%	no
S-no-majority	0.2506	0.3135	0.5447	21%	10%	no
S-contested (pooled)	0.3902	0.4575	0.6700	24%	29%	no

Why these two quantities settle the question

the modal label of the same three references is the best any single-label predictor can achieve against them. It bounds the comparison. A model that merely reaches it is doing what any modal-vote rule does; only a model that EXCEEDS it is doing something an aggregation rule could not. the fraction of images on which the held-out annotator's label is shared by none of the other three. Such an annotator cannot score well whatever their skill. It is fixed by the stratum definition: 0% where all four agree, 25% on a 3-1 stratum, 50% on a 2-1-1 stratum. Where it is high, the human side is structurally handicapped and the contrast is not a like-for-like skill test.



Each held-out annotator is scored against the other three, and the model is scored against the same three, on the same images and the same patient resample. On the unanimous stratum the human side is 1.0 by construction, so that contrast is greyed out. The triangle is the modal-vote oracle — the best any single-label predictor can do against the same three references. It bounds the comparison: the model chooses a label, an annotator is stuck with theirs. On contested images the model recovers only 24% of that headroom, so it out-predicts an individual annotator without reaching the panel's own modal vote.

Figure 1. P6-A. Left: held-out annotator, model and the modal-vote oracle on a common axis; the unanimous stratum is greyed because the human side is 1.0 by construction there. Right: the paired difference with patient-clustered 95% intervals.

Table 8. The pre-registered verdicts stand as the frozen rules produced them. These are the claims the evidence supports once P6-AMD-5 is accounted for, and they are the form used everywhere else in this thesis.

Stratum	What may actually be claimed
S-unanimous	NOT A COMPARISON — the human side is 1.0 by construction on a stratum defined by unanimity.
S-majority	OUT-PREDICTS AN INDIVIDUAL ANNOTATOR, BUT NOT THE PANEL'S OWN MODAL VOTE. The model recovers 16% of the headroom between a held-out annotator (0.4893) and the modal-vote oracle (0.7423). The pre-registered 'INDISTINGUISHABLE FROM THE HUMAN

Stratum	What may actually be claimed
	PANEL' therefore reflects that predicting a panel is an easier task than being a member of it — the model chooses a label, an annotator is stuck with theirs, and 25% of held-out annotators on this stratum are singletons who cannot score well whatever their skill. It is NOT evidence of superior anatomical judgement, and a substantial model shortfall against the attainable ceiling remains.
S-plurality	OUT-PREDICTS AN INDIVIDUAL ANNOTATOR, BUT NOT THE PANEL'S OWN MODAL VOTE. The model recovers 40% of the headroom between a held-out annotator (0.1436) and the modal-vote oracle (0.4492). The pre-registered 'ABOVE THE HUMAN PANEL' therefore reflects that predicting a panel is an easier task than being a member of it — the model chooses a label, an annotator is stuck with theirs, and 50% of held-out annotators on this stratum are singletons who cannot score well whatever their skill. It is NOT evidence of superior anatomical judgement, and a substantial model shortfall against the attainable ceiling remains.
S-no-majority	OUT-PREDICTS AN INDIVIDUAL ANNOTATOR, BUT NOT THE PANEL'S OWN MODAL VOTE. The model recovers 21% of the headroom between a held-out annotator (0.2506) and the modal-vote oracle (0.5447). The pre-registered 'ABOVE THE HUMAN PANEL' therefore reflects that predicting a panel is an easier task than being a member of it — the model chooses a label, an annotator is stuck with theirs, and 10% of held-out annotators on this stratum are singletons who cannot score well whatever their skill. It is NOT evidence of superior anatomical judgement, and a substantial model shortfall against the attainable ceiling remains.
S-contested (pooled)	OUT-PREDICTS AN INDIVIDUAL ANNOTATOR, BUT NOT THE PANEL'S OWN MODAL VOTE. The model recovers 24% of the headroom between a held-out annotator (0.3902) and the modal-vote oracle (0.6700). The pre-registered 'ABOVE THE HUMAN PANEL' therefore reflects that predicting a panel is an easier task than being a member of it — the model chooses a label, an annotator is stuck with theirs, and 29% of held-out annotators on this stratum are singletons who cannot score well whatever their skill. It is NOT evidence of superior anatomical judgement, and a substantial model shortfall against the attainable ceiling remains.

The pre-registered interpretation, and why it is only partly licensed

The pre-registration fixed the interpretation 'if the model is INDISTINGUISHABLE from the human panel on the contested strata while scoring 26-49 macro F1, the low absolute scores are a property of the task and not of the model.' That inference is now only PARTLY licensed. The modal-vote oracle added in P6-AMD-5 shows the attainable ceiling on contested images is 0.67, not 1.0, so a large part of the Phase 3 decline IS the ceiling moving -- consistent with the Phase 3B ceiling-normalised analysis. But the model reaches only about a quarter of the way from an individual annotator to that ceiling, so a substantial model shortfall remains and 'the low scores are a property of the task' overstates it. The thesis must use the qualified_verdict field, not verdict_summary.

3.2 P6-B: confusion geometry with intervals on both sides

Table 9. P6-B. Adjacent-wall and neighbouring-station shares, model and human, measured on the same images. Arm C2.

Stratum	Wall model %	Wall human %	Δ	Station model %	Station human %	Δ
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Stratum	Wal l mo del %	Wal l hu ma n %	Δ	Stat ion mo del %	Stat ion hu ma n %	Δ
S-unanimous	96.25	undefined	—	86.98	undefined	—
S-majority	93.02	92.93	-0.74	88.59	97.92	-10.80
S-plurality	85.85	86.15	-0.15	84.54	90.05	-4.56
S-no-majority	90.94	93.63	-2.55	87.21	95.65	-8.56
S-contested (pooled)	90.59	91.10	-0.92	87.21	94.84	-8.19

X3, settled

X3 is settled, and by a stronger argument than the one that raised it. The amendment withdrew the Phase 3 claim because a model INTERVAL was compared against a human POINT estimate. Measuring both sides on the same images shows something more basic: S-unanimous is defined by all four annotators agreeing, so it contains ZERO annotator disagreement events and the human error geometry is UNDEFINED there -- not imprecise, undefined. The Phase 3 comparison therefore set the model's geometry on the unanimous stratum against a human benchmark computed over the whole corpus, which is dominated by contested images. It was a cross-population comparison, and neither the 0.12-point 'match' nor the 7.5-point 'shortfall' was ever a like-for-like quantity.

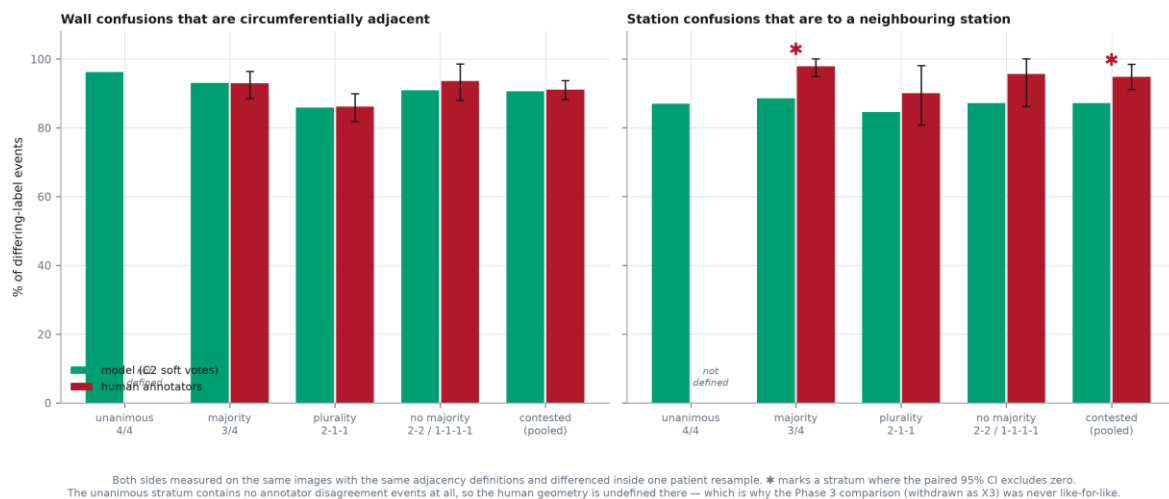


Figure 2. P6-B. Both sides carry a patient-clustered interval and are differenced inside one resample. * marks a stratum whose paired interval excludes zero.

3.3 P6-C: attribution

P6-AMD-4 — the pre-registered primary is not estimable

the pre-registered primary endpoint is NOT ESTIMABLE on its pre-registered stratum, for a structural reason the pre-registration did not anticipate. annotator vote entropy is a deterministic function of the vote PATTERN -- a 3-1 split is always 0.5623 nats, a 2-1-1 split always 1.0397 -- and the agreement strata are DEFINED by that same pattern. Vote entropy is therefore constant within S-unanimous, S-majority and S-plurality alike (1 distinct value each), and takes only 2 values on S-no-majority. A within-stratum correlation against a constant does not exist. P6-C1 is

reported as NOT ESTIMABLE rather than as a null. Reporting it as 'no association' would assert a measurement that was never possible.

Table 10. P6-C1 on S-majority. The pre-registered signal (vote entropy) is constant within the stratum and yields no estimate. The right-hand columns are the declared-exploratory substitute — anatomical vote spread — which varies within a tier. They are exploratory and are not a pre-registered endpoint.

Configurati on	CA M dis per sion	p vs vot e ent rop y	Pre- regist ered verdi ct	p vs vot e spr ead	95% CI	Expl orat ory verdi ct
C0 hard label, 4/4 cohort (Phase 2 reference)	0.6 723	n/a	NOT ESTIM ABLE (vote entro py is consta nt on this stratu m by constr uction)	- 0.0 099	[- 0.107 2, 0.095 8]	NO ASSO CIATI ON
C1 hard majority label, extended cohort	0.6 831	n/a	NOT ESTIM ABLE (vote entro py is consta nt on this stratu m by constr uction)	- 0.0 310	[- 0.134 8, 0.084 5]	NO ASSO CIATI ON
C2 vote proportions	0.6 320	n/a	NOT ESTIM ABLE (vote entro py is consta nt on this stratu m by constr uction)	0.0 358	[- 0.065 4, 0.149 7]	NO ASSO CIATI ON
C3 hard label + matched label smoothing (control)	0.6 142	n/a	NOT ESTIM ABLE (vote entro py is consta nt on this stratu m by constr uction)	0.1 062	[- 0.006 8, 0.233 8]	NO ASSO CIATI ON
C4 vote proportions + anatomical penalty	0.6 282	n/a	NOT ESTIM ABLE (vote entro py is consta nt on this stratu m by constr uction)	0.0 294	[- 0.087 5, 0.151 9]	NO ASSO CIATI ON

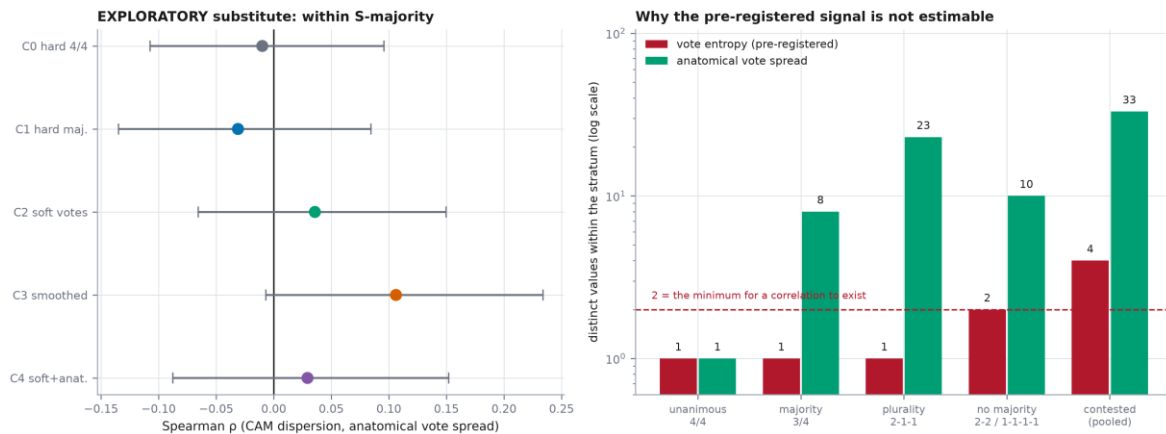
Configuration	CAM dispersion	ρ vs vote entropy	Pre-registered verdict	ρ vs vote spread	95% CI	Exploratory verdict
			not on this stratum by construction)			

What the substitute signal is

a declared EXPLORATORY substitute is reported alongside: the mean pairwise ANATOMICAL distance between the four annotators' labels under the frozen Phase 4 distance matrix. It carries the same disagreement information but distinguishes a dissenter one wall away from a dissenter at the far end of the stomach, so it varies within a tier (8, 23 and 10 distinct values on S-majority, S-plurality and S-no-majority). It is post-hoc and is labelled exploratory everywhere it appears; it is NOT a substitute pre-registered endpoint.

A consequence for Phase 4's RQ3, carried back

this also bears on Phase 4's RQ3, which reported 'within-stratum' entropy correlations of 0.02-0.08. On this corpus that quantity can only be non-degenerate where a stratum mixes vote patterns. Phase 7 should restate the RQ3 finding with this structural limit made explicit.



The pre-registered primary signal, annotator vote entropy, is NOT ESTIMABLE within a stratum: entropy is a deterministic function of the vote pattern and the strata are defined by that pattern.
A 3-1 vote split always has entropy 0.5623 nats and a 2-1-1 split always 1.0397, so vote entropy is constant inside every stratum defined by a single vote pattern. Anatomical vote spread carries the same disagreement information but separates a dissenter one wall away from one at the far end of the stomach, so it varies within a tier. The substitute is post-hoc and labelled exploratory.

Figure 3. P6-C1. Left: the primary within-stratum correlation per arm. Right: the pooled value against the within-stratum value, showing the size of the stratum-membership confound.

Table 11. P6-C2, secondary: inter-seed overlap of the top-20% attribution masks.

Configuration	IoU unanimous	IoU contested	Δ	95% CI	Verdict
C0 hard label, 4/4 cohort (Phase 2 reference)	0.4856	0.4599	+0.0257	[0.0079, 0.0450]	ATTRIBUTION DESTABILISES ON CONTESTED IMAGES
C1 hard majority label,	0.50	0.48	+0.	[0.0053,	ATTRIBUTION

Configuration	IoU unanimous	IoU contested	Δ	95% CI	Verdict
extended cohort	33	24	0.209	0.0372]	DESTABILISES ON CONTESTED IMAGES
C2 vote proportions	0.6759	0.6525	+0.0233	[0.0030, 0.0413]	ATTRIBUTE DESTABILISES ON CONTESTED IMAGES
C3 hard label + matched label smoothing (control)	0.6783	0.6230	+0.0552	[0.0347, 0.0802]	ATTRIBUTE DESTABILISES ON CONTESTED IMAGES
C4 vote proportions + anatomical penalty	0.6703	0.6307	+0.0396	[0.0190, 0.0617]	ATTRIBUTE DESTABILISES ON CONTESTED IMAGES

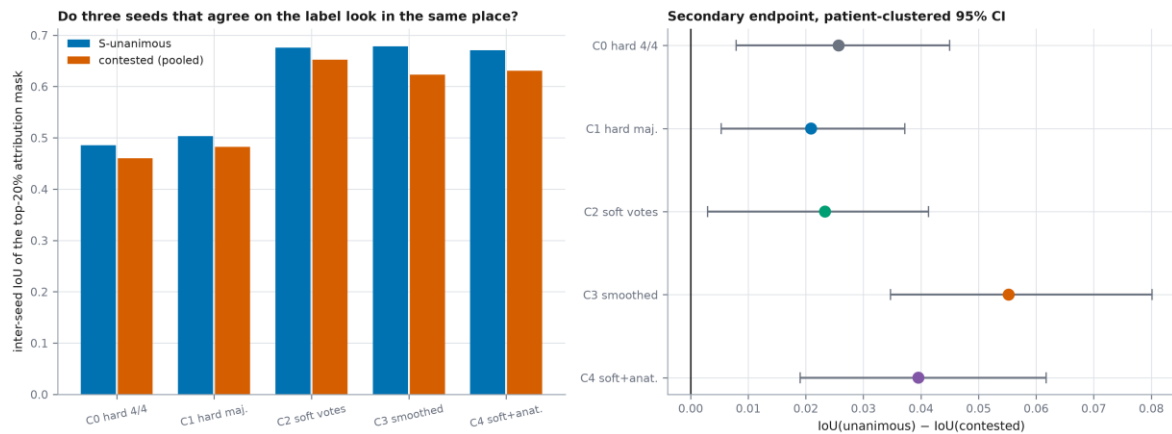
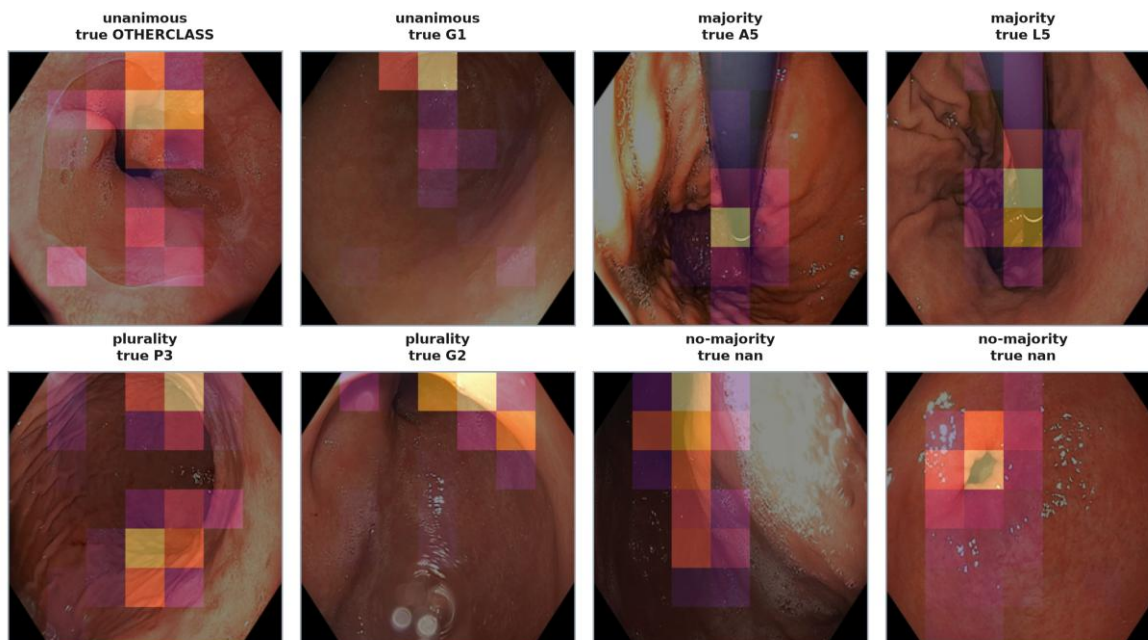


Figure 4. P6-C2. Whether three seeds that agree on the label also agree on where the evidence is.



Grad-CAM for the committed prediction, arm C2 seed 1. These panels are illustrative only. The quantitative endpoint is Figure F40: within S-majority, $p = \text{None}$ (NOT ESTIMABLE (vote entropy is constant on this stratum by construction)).

Figure 5. Qualitative Grad-CAM panels, shown after the quantitative endpoint and captioned with it. These illustrate; Figure F40 measures.

3.4 P6-D: selective prediction

Table 12. P6-D, Internal — GastroHUN test split. AUROC lower is better.

Configuration	AU RC	95% CI	Cover age at 10% risk	Risk at 80% cover age
C0 hard label, 4/4 cohort (Phase 2 reference)	0.0977	[0.0826, 0.1296]	0.575	0.180
C1 hard majority label, extended cohort	0.0927	[0.0696, 0.1108]	0.592	0.174
C2 vote proportions	0.0828	[0.0620, 0.1017]	0.647	0.156
C3 hard label + matched label smoothing (control)	0.0822	[0.0623, 0.1004]	0.652	0.156
C4 vote proportions + anatomical penalty	0.0870	[0.0646, 0.1045]	0.635	0.162

Table 13. P6-D, External — HyperKvasir + GastroVision. AUROC lower is better.

Configuration	AU RC	95% CI	Cover age at 10% risk	Risk at 80% cover age
C0 hard label, 4/4 cohort (Phase 2 reference)	0.3315	—	0.112	0.465
C1 hard majority	0.3	—	0.085	0.463

Configuration	AU RC	95% CI	Cover age at 10% risk	Risk at 80% cover age
label, extended cohort	382			
C2 vote proportions	0.1906	—	0.230	0.286
C3 hard label + matched label smoothing (control)	0.3235	—	0.051	0.443
C4 vote proportions + anatomical penalty	0.2319	—	0.171	0.342

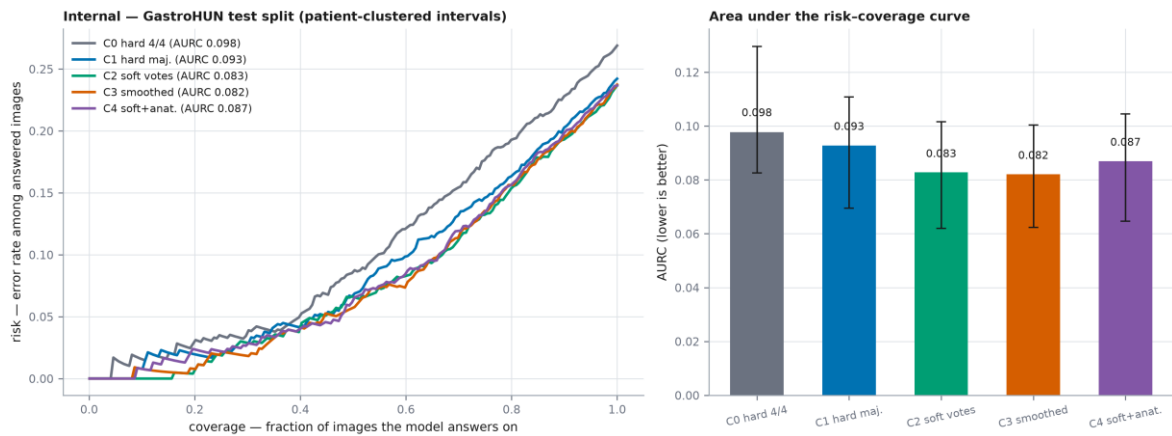


Figure 6. P6-D internal. Risk-coverage curves and AURC with patient-clustered intervals.

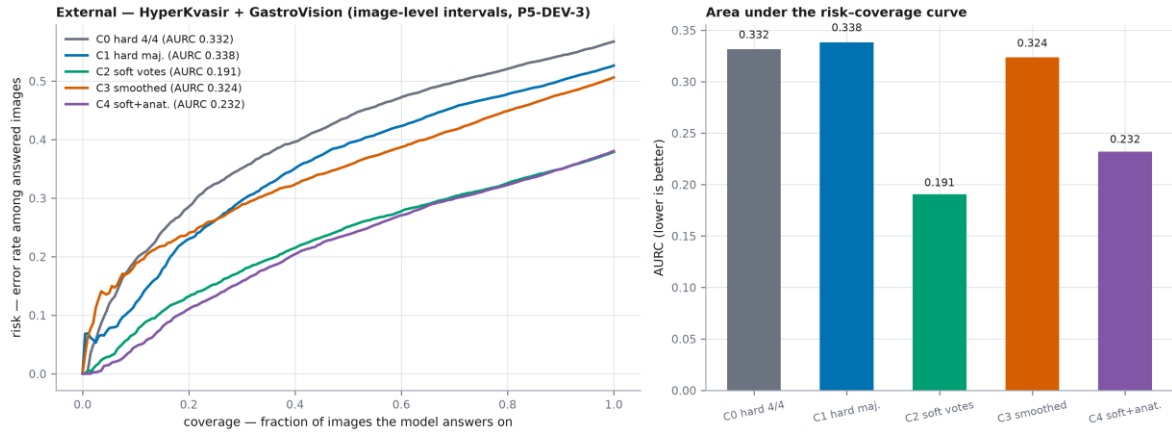


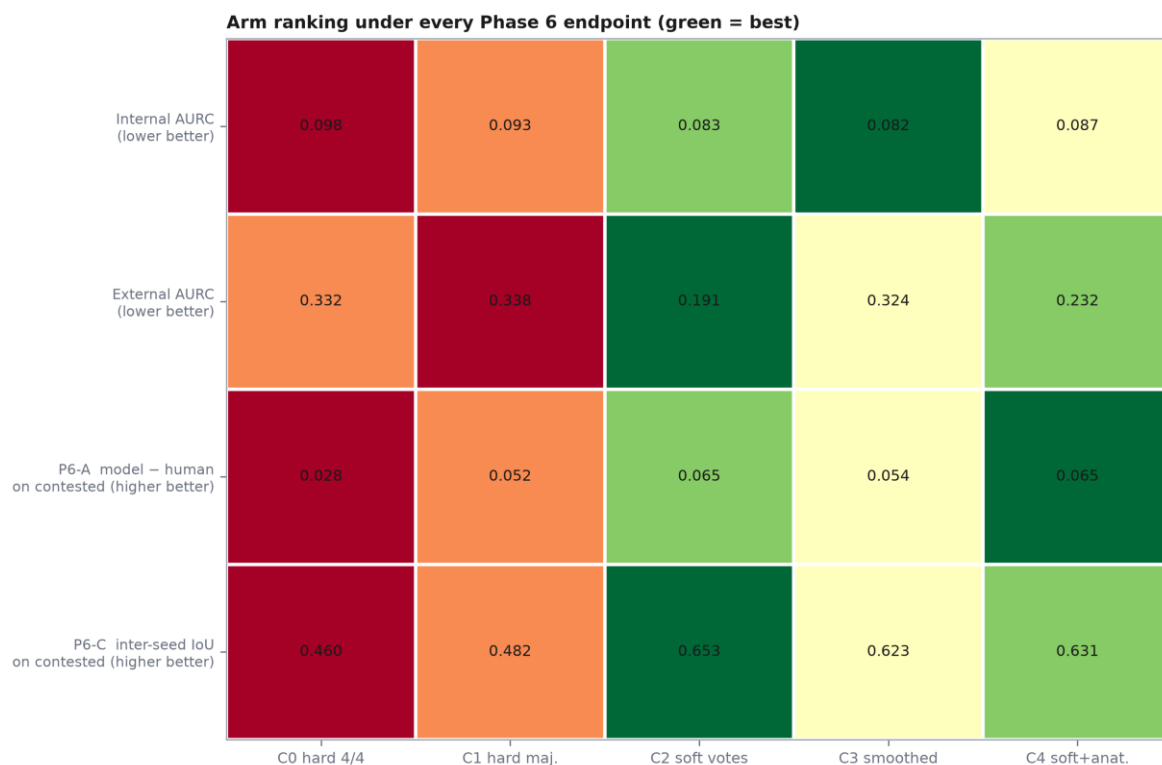
Figure 7. P6-D external. The same measurement on the Phase 5 panel, where rejecting an out-of-protocol image is the correct action. Intervals here are image-level (P5-DEV-3) and are not comparable with the internal ones.

Does the single-operating-point result generalise?

lower AURC is better and higher rejection is better, so the correlated quantities are $(-AURC)$ and the Phase 5 rejection rate; a POSITIVE rho means the two endpoints agree about which arms are good Measured $p = 1.0$, 95% CI $[0.80, 0.80] \rightarrow$ CONSISTENT WITH THE PHASE 5 RANKING. with only five arms the Spearman statistic takes a small number of discrete values, so this interval is coarse by construction and is reported as a consistency check rather than as a powered test. The bootstrap interval does not contain the point estimate. This is not an error: Spearman's rho on five items can take only a handful of values, the full-sample ordering achieves the maximum 1.0, and every resample that perturbs one adjacent pair drops it to 0.9 or 0.8. The percentile interval therefore sits entirely below a point estimate that is at the boundary of the statistic's range. Read the interval as 'the ordering agrees strongly

under resampling', not as a conventional confidence interval.

3.5 Cross-endpoint synthesis



Cell shading is the rank within the row; the printed value is the measured quantity. Rows disagree, which is the point: no single arm wins everywhere.

Figure 8. Arm ranking under every Phase 6 endpoint. The rows disagree, which is the finding: no configuration wins everywhere, and the endpoint that separates them most is external selective prediction.

4. Discussion

4.1 Ambiguity or capacity: what P6-A and P6-B jointly license

The pre-registration fixed the interpretation of P6-A before the numbers existed: if the model is indistinguishable from the human panel on the contested strata while scoring in the twenties and forties, the low absolute scores are a property of the task. The frozen rule returned above the human panel on the pooled contested stratum. Taken at face value that is a striking claim, and it is the reason this section does not take it at face value.

Two asymmetries were found on re-reading the construction, and both were then measured. First, exposure: the model is trained on targets derived from this panel and is optimised to predict its consensus, while the held-out annotator is simply being themselves. Second, and more decisively, choice: the model selects a label, whereas the annotator is stuck with the one they gave. The modal vote of the same three references — the best any single-label predictor can achieve against them — scores 0.6700, against 0.3902 for the held-out annotator. The model reaches 0.4575, which is 24% of that headroom. It does not exceed the oracle on any stratum.

The defensible reading is therefore narrower than the frozen rule suggests, and more useful. Predicting a panel is an easier task than being a member of one, and on the 2-1-1 stratum half of all held-out annotators are singletons who cannot score well whatever their skill. What P6-A establishes is that the attainable ceiling on contested images is 0.67 rather than 1.00 — so a large part of the Phase 3 decline is the ceiling moving, exactly as the Phase 3B ceiling-normalised analysis found — while a substantial model shortfall against that reduced ceiling remains. The decline is neither purely an artefact of the reference standard nor purely a failure of the classifier, and the contribution of this endpoint is to separate the two and size them.

P6-B sharpens this. The wall axis mirrors human geometry on every stratum where the comparison is defined: when the model confuses the circumferential direction of the scope, it confuses it the way endoscopists do. The station axis does not. The model's station errors travel further along the insertion axis than human disagreements do, and that difference survives a patient-clustered interval on the pooled contested stratum. That is a specific, actionable deficit rather than a general one.

4.2 What attribution did and did not explain

P6-C1 did not return a null; it returned a discovery about the measurement itself. Annotator vote entropy is a deterministic function of the vote pattern, and the agreement strata are defined by that pattern, so entropy is constant inside S-unanimous, S-majority and S-plurality alike. A within-stratum correlation against it cannot exist. Reporting this as 'no association' would have asserted a measurement that was never available, so it is reported as NOT ESTIMABLE.

The declared-exploratory substitute — the mean pairwise anatomical distance between the four annotators' labels, which does vary within a tier — returns NO ASSOCIATION ($p = 0.0358$, 95% CI [-0.0654, 0.1497]). Taken with Phase 4's finding on predictive entropy, the picture is consistent: this model's uncertainty is not legible either in its output distribution or in where it looks.

P6-C2 is the positive result of this endpoint. Attribution destabilises on contested images for every arm, and the three soft-target arms are markedly more spatially consistent across seeds than the two hard-label arms (0.676 for C2 against 0.486 for C0 on unanimous images). Training on the vote distribution does not make the model's confidence track disagreement, but it does make three independently seeded models agree about where the evidence is.

All of this is evidence about Grad-CAM specifically. No other attribution method was run, by design (P6-DEV-3), so that a method could not be selected after seeing which one correlated best.

4.3 Selective prediction as the deployable form of the calibration finding

Internally the arms are nearly indistinguishable on AURC, which is consistent with every other internal endpoint in this project failing to separate them. Externally they separate sharply: C2 attains an AURC of 0.1906 while the internally-best arm C3 reaches only 0.3235. This is the Phase 4 calibration finding in deployable form: the question is not which arm is most accurate but which arm's confidence can be trusted to decide when not to answer.

4.4 Limitations specific to this phase

- Grad-CAM is one attribution method among several, and the P6-C verdict is evidence about it alone (P6-DEV-3).
- Four annotators bound the precision of the human comparator; the held-out construction leaves a reference panel of three.
- The human comparator is degenerate on any stratum defined by unanimity, and the human error geometry does not exist there at all. Both are declared rather than worked around.
- Contested strata remain small ($n = 342, 127$ and 81), so the per-stratum intervals are wide and the pooled stratum carries the endpoints.
- External intervals are image-level, inheriting the Phase 5 declaration P5-DEV-3, and must not be compared against the patient-clustered intervals in the same report.
- The strata are defined by the same annotator agreement that determines the human comparator's score, so the human curve is partly mechanical. This does not affect the model-minus-human contrast, which is measured on identical rows, but it does mean the human curve alone should not be read as a free-standing estimate of expert accuracy.

5. Conclusion

5.1 Answers to the four endpoints

Table 14. Every verdict selected by a rule frozen before scoring.

Endpoint	Verdict
P6-A S-majority	INDISTINGUISHABLE FROM THE HUMAN PANEL — qualified: OUT-PREDICTS AN INDIVIDUAL ANNOTATOR, BUT NOT THE PANEL'S OWN MODAL VOTE.
P6-A S-plurality	ABOVE THE HUMAN PANEL — qualified: OUT-PREDICTS AN INDIVIDUAL ANNOTATOR, BUT NOT THE PANEL'S OWN MODAL VOTE.
P6-A S-no-majority	ABOVE THE HUMAN PANEL — qualified: OUT-PREDICTS AN INDIVIDUAL ANNOTATOR, BUT NOT THE PANEL'S OWN MODAL VOTE.
P6-A S-contested (pooled)	ABOVE THE HUMAN PANEL — qualified: OUT-PREDICTS AN INDIVIDUAL ANNOTATOR, BUT NOT THE PANEL'S OWN MODAL VOTE.
P6-B wall axis, contested	MIRRORS HUMAN GEOMETRY
P6-B station axis, contested	DIVERGES FROM HUMAN GEOMETRY (model less adjacent)
P6-C1 attribution vs disagreement (pre-registered)	NOT ESTIMABLE (vote entropy is constant on this stratum by construction)
P6-C1b same, exploratory substitute signal	NO ASSOCIATION
P6-C2 attribution stability	ATTRIBUTION DESTABILISES ON CONTESTED IMAGES
P6-D vs the Phase 5 ranking	CONSISTENT WITH THE PHASE 5 RANKING

5.2 Carry-forward to Phase 7

- The thesis's central claim is that agreement stratification separates a falling reference standard from a falling classifier, and that the literature reports them as one quantity. Both fall; the ceiling accounts for the larger share, and a real model shortfall remains.
- Never state P6-A as 'the model beats the expert'. State it as the qualified verdict: the model out-predicts an individual annotator but recovers only about a quarter of the headroom to the panel's own modal vote, and it exceeds that oracle on no stratum.

- Report the human comparator curve AND the modal-vote oracle alongside every stratified performance figure. Phase 3's numbers mislead without the first, and the first misleads without the second.
- The station axis is the specific residual deficit worth naming; the wall axis is not.
- Use external AURC as the arm-selection endpoint. It is the only measurement in the project that separates the configurations decisively and it agrees with the Phase 5 rejection ranking.

Appendix A. Per-held-out-annotator breakdown

Table 15. Per-held-out-annotator scores on S-majority. FG2 is the outlier annotator identified in Phase 0 and is included unchanged.

Held out	Reference panel	Human	Model (C2)	Δ
G1	G2, FG1, FG2	0.5072	0.5436	+0.0365
G2	G1, FG1, FG2	0.3806	0.5270	+0.1463
FG1	G1, G2, FG2	0.5514	0.5157	-0.0358
FG2	G1, G2, FG1	0.5180	0.5340	+0.0160

Table 16. Per-held-out-annotator scores on S-contested (pooled). FG2 is the outlier annotator identified in Phase 0 and is included unchanged.

Held out	Reference panel	Human	Model (C2)	Δ
G1	G2, FG1, FG2	0.4018	0.4705	+0.0687
G2	G1, FG1, FG2	0.3117	0.4510	+0.1394
FG1	G1, G2, FG2	0.4416	0.4476	+0.0061
FG2	G1, G2, FG1	0.4058	0.4609	+0.0551

Appendix B. Pre-registration record and declared deviations

Table 17. Deviations declared in the frozen pre-registration.

ID	Blueprint item	Adopted	Because
P6 - DEV-1	blueprint sec.4 PHASE 6 says 'Grad-CAM on contested images'	Grad-CAM on ALL 1,353 test images	the contested strata cannot be interpreted without the unanimous stratum as a reference level, and the secondary endpoint P6-C2 is defined as a difference between them. Computing the unanimous maps is cheap and refusing to compute them would make the contrast impossible.
P6 - DEV-	blueprint sec.4 PHASE 6 frames the	P6-A (human comparator) and P6-	the blueprint's own stated purpose for the phase is to decide whether residual error

ID	Blueprint item	Adopted	Because
2	phase as a confusion-matrix comparison only	D (selective prediction) added	reflects 'genuine visual ambiguity rather than model capacity'. Confusion geometry shows the errors have the same SHAPE, not the same DIFFICULTY; only a human comparator can license the ambiguity claim. P6-D discharges the Phase 5 carry-forward, which the blueprint assigns to Phases 6-7 but does not specify.
P6 - DE V-3	attribution method sweep	Grad-CAM only	a sweep over Grad-CAM++/ScoreCAM/IntegratedGradients would quadruple the backward passes and, more importantly, invite selection of whichever method produced the strongest correlation. One method, pre-registered, reported whatever it says.

Table 18. Amendments. Each was forced by a gate or by a numerical fact the pre-registration did not anticipate, and each is recorded rather than silently adopted.

ID	Amendment discovered during execution
P6-AM D-1	P6-AMD-1: the pre-registration did not anticipate that the human side of P6-A is 1.0 by construction on any stratum defined by unanimity. The S-unanimous verdict is therefore reported as UNINFORMATIVE (BY CONSTRUCTION) rather than as BELOW THE HUMAN PANEL. The contested strata, where annotators genuinely disagree, are unaffected and carry the endpoint.
P6-AM D-2	P6-AMD-2: the pre-registration assumed the human geometry would be estimable on every stratum. It is not estimable on S-unanimous, for a structural reason. The S-unanimous row is reported as NOT DEFINED rather than as a verdict, and the endpoint is carried by the contested strata.
P6-AM D-3	P6-AMD-3: the pre-registration says the CAM target is 'the model's own predicted class'. That phrase is ambiguous between the live argmax and the committed one; it is resolved here in favour of the committed prediction, because that is the decision every other Phase 3-6 endpoint scores. The resolution was forced by gate P6.4b firing on 1 of 1,353 images, and is recorded rather than silently adopted.
P6-AM D-4	the pre-registered primary endpoint is NOT ESTIMABLE on its pre-registered stratum, for a structural reason the pre-registration did not anticipate. annotator vote entropy is a deterministic function of the vote PATTERN -- a 3-1 split is always 0.5623 nats, a 2-1-1 split always 1.0397 -- and the agreement strata are DEFINED by that same pattern. Vote entropy is therefore constant within S-unanimous, S-majority and S-plurality alike (1 distinct value each), and takes only 2 values on S-no-majority. A within-stratum correlation against a constant does not exist. P6-C1 is reported as NOT ESTIMABLE rather than as a null. Reporting it as

ID	Amendment discovered during execution
	'no association' would assert a measurement that was never possible. a declared EXPLORATORY substitute is reported alongside: the mean pairwise ANATOMICAL distance between the four annotators' labels under the frozen Phase 4 distance matrix. It carries the same disagreement information but distinguishes a dissenter one wall away from a dissenter at the far end of the stomach, so it varies within a tier (8, 23 and 10 distinct values on S-majority, S-plurality and S-no-majority). It is post-hoc and is labelled exploratory everywhere it appears; it is NOT a substitute pre-registered endpoint.

Appendix C. Script and artefact manifest

Table 19. Reproducibility index. Running the scripts in this order regenerates every number, figure and verdict in this report.

Script	Artefact	Step
src/models/phase6_prereg.py	reports/phase6_prereg.json	P6.0
src/models/phase6_common.py	(shared primitives, gates P6.1a/b)	P6.1
src/models/phase6_human.py	reports/phase6_human.json	P6.2
src/models/phase6_geometry.py	reports/phase6_geometry.json	P6.3
src/models/phase6_cam.py	reports/phase6_cams_*.npz, phase6_cam_gate.json	P6.4
src/models/phase6_cam_eval.py	reports/phase6_cam_eval.json	P6.5
src/models/phase6_selective.py	reports/phase6_selective.json	P6.6
src/report/figures_phase6.py	figures_phase6/P6_F38–F45	P6.7
src/report/build_phase6_docx.py	Phase6_Report.docx	P6.8

Appendix D. Analyses NOT executed, and what each would cost

- An attribution-method sweep (Grad-CAM++, ScoreCAM, Integrated Gradients). Roughly 3× the backward passes, about 45 minutes on this hardware — declined because it would invite selecting whichever method correlated best.
- A second backbone, to test whether the calibration and attribution findings are ConvNeXt-specific. Nine training runs, roughly 10 GPU hours. Not executed; the strongest remaining threat to external validity of the Phase 6 conclusions.
- Human comparator on the external corpora. Not possible: neither corpus publishes per-annotator labels.
- A patient-clustered external interval. Not possible: neither corpus publishes a case identifier.